Jing-Yi Lu

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Field Museum of Natural History - Chicago, IL, USA

Education

- **University of Chicago** *2018 - 2025*
  Ph.D. in Evolutionary Biology USA
  - Dissertation: Pollinator Shifts, Floral Evolution, and Speciation in *Aeschynanthus* (Gesneriaceae)
  - Advisor: Dr. Richard H. Ree

- **University of Chicago** *2018 - 2022*
  M.Sc. in Evolutionary Biology USA

- **National Taiwan University** *2013 - 2018*
  B.Sc. in Life Science Taiwan

- **University of Melbourne** *2016 - 2017*
  Exchange student Australia

Publications

*Co-first authors: contribute equally to this research

Published

[1] **J-Y Lu**, YW Xing, H T Luu, RH Ree. 2026. Testing macroevolutionary predictions of the Grant-Stebbins model in the origin of *Aeschynanthus acuminatus*. *New Phytol.* 249(6):3137-3148. DOI: 10.1111/nph.70871

[2] C-C Yu*, Z-Q Mo*, **J-Y Lu**, J-J Hu, J-B Yang, S-F Li, L-B Jia. 2025. Revisiting the Molecular Circumscription of Sladeniaceae sensu lato (Ericales): Insights from Phylogenetic Discordance. *Am. J. Bot.* 112:e70141. DOI: 10.1002/ajb2.70141

[3] K-S Chen*, **J-Y Lu***, C-N Wang. 2019. Effective pollination of *Aeschynanthus acuminatus* (Gesneriaceae) by generalist passerines, in sunbird-absent East Asia. *Sci. Rep.* 9:17552. DOI: 10.1038/s41598-019-53035-2.

[4] K-T Hsin*, **J-Y Lu***, M Möller, C-N Wang. 2019. Gene duplication and relaxation from selective constraints of *GCYC* genes correlated with various floral symmetry patterns in Asiatic Gesneriaceae tribe Trichosporeae. *PloS one* 14(1):e0210054-e0210054. DOI: 10.1371/journal.pone.0210054

[5] H-J Hsu, C-W He, W-H Kuo, K-T Hsin, **J-Y Lu**, Z-J Pan, C-N Wang. 2018. Genetic analysis of floral symmetry transition in African violet suggest involvement of trans-acting factor for *CYCLOIDEA* expression shifts. *Front. Plant Sci.* 9:1008. DOI: 10.3389/fpls.2018.01008

In Press

[1] **J-Y Lu***, S-J Ling*, M-X Ren, RH Ree. Accepted. Sunbird pollination of *Aeschynanthus moningerae* and its implications for floral adaptations for hovering. *Am. Nat.* DOI: 10.1086/744314

[2] Q Li, K Hu, J Wang, RS Fuller, **J-Y Lu**, H Wang, J Liu, RH Ree, X Kou. 2026. Genome-wide signatures and ecological strategies show the adaptation of an alpine *Rhododendron* across elevations. *Mol. Ecol.* 35(18): e70542. DOI: 10.1111/mec.70542

[3] Q Shen, N He, Y Kang, F Wang, **J-Y Lu**, Q Li. 2026. Multidimensional floral disparity in *Rhododendron*: Evolutionary constraints and environmental filtering in a generalized pollination system. *Plant Divers.* DOI: 10.1016/j.pld.2026.07.003

In Preparation

[1] **J-Y Lu**, YW Xing, H T Luu, M-X Ren, S-J Ling, RH Ree, In Prep. Pollinator-driven divergence and speciation in three *Aeschynanthus* species.

[2] C-C Yu, Y-L Huang, T Ito, T-T He, K-F Chung, **J-Y Lu**. In Prep. The origin of endemic diversity on a geologically young continental island: multiple origins or in situ diversification of *Berberis* in Taiwan.

Grants and Awards

- **Armour Graduate Fellowship** *2022-2023*
  Field Museum of Natural History, USA
- **Negaunee fieldwork fund | P.I.: Dr. R.H. Ree** *2022*
  Field Museum of Natural History, USA
- **Fund for Research in Pritzker Lab (Field Museum)** *2021*
  Committee on Evolutionary Biology, University of Chicago, USA
- **Henry Hinds Fund for Graduate Student Research in Evolutionary Biology** *2021*
  Committee on Evolutionary Biology, University of Chicago, USA
- **Nellie D. Sleeth Scholarships** *2021*
  The Gesneriad Society, USA
- **Negaunee Fieldwork Fund | P.I.: Dr. R.H. Ree** *2020*
  Field Museum of Natural History, USA
- **Oral Presentation Award** *2018*
  Congress of Animal Behavior and Ecology, Taiwan
- **College Student Research Creativity Award** *2017*
  Ministry of Science and Technology, Taiwan
- **Outstanding Student Award** *2017*
  XIX International Botanical Congress, China
- **The Dean Award** *2017*
  College of Life Science, National Taiwan University, Taiwan
- **Scholarship of Overseas Education** *2017*
  College of Life Science, National Taiwan University, Taiwan
- **Grant for Undergraduate Research | MOST 105-2815-C-002-099-B** *2016-2017*
  Ministry of Science and Technology, Taiwan
- **Undergraduate Travel Award** *2016*
  Society of Molecular Biology and Evolution Conference, Australia
- **Mr. Cheng, Yu-Bin Memorial Scholarship | for research in plant systematics and phylogenetics** *2016*
  College of Life Science, National Taiwan University, Taiwan

Teaching Experience

- **Teaching Assistant | Reconstructing the Tree of Life: An Introduction to Phylogenetics** *2021*
  University of Chicago, USA
- **Teaching Assistant Training** *2020*
  University of Chicago, USA

Conference Presentations

Oral Presentation

[1] **J-Y Lu**, YW Xing, H T Luu, M-X Ren, S-J Ling, RH Ree. Pollinator-mediated speciation on two continental islands: insights from bird-pollinated *Aeschynanthus*. Island Biology 2026: 2-6 Nov, Atami City, Japan.

[2] **J-Y Lu**, YW Xing, H T Luu, M-X Ren, S-J Ling, RH Ree. Pollinator-driven divergence and speciation in three *Aeschynanthus* species. Botany 2025: 26-30 July, Palm Springs, CA, USA.

[3] **J-Y Lu**, YW Xing, H T Luu, M-X Ren, S-J Ling, RH Ree. Pollinator shifts, floral evolution, and geographic range evolution in the genetic divergence of three *Aeschynanthus* species. XX IBC International Botanical Congress: 21-27 July 2024, Madrid, Spain.

[4] **J-Y Lu**, YW Xing, H T Luu, RH Ree. Pollinator shifts and geographic range evolution in the origin of *Aeschynanthus acuminatus* (Gesneriaceae). Botany 2023: 22-26 July, Boise, ID, USA.

[5] **J-Y Lu**, S-J Ling, M-X Ren, RH Ree. 2022. Floral traits limit pollinator assemblages and determine the visiting behavior of nectar specialist passerines. Botany 2022: 24-27 July, Anchorage, AK, USA.

[6] **J-Y Lu**, K-S Chen, C-N Wang. 2018. Generalist passerines as specialized pollinators in the absence of sunbirds: Highly efficient generalized passerine pollination system of East Asian *Aeschynanthus acuminatus* (Gesneriaceae). Congress of Animal Behavior and Ecology: 22-23 January, Hsinchu, Taiwan.

Poster Presentation

[1] **J-Y Lu**, YW Xing, H T Luu, M-X Ren, S-J Ling, RH Ree. Pollinator shifts, floral evolution, and geographic range evolution in the genetic divergence of three *Aeschynanthus* species. Chicago Plant Science Symposium 2024: 20 April 2024, Glencoe, IL, USA.

[2] **J-Y Lu**, K-S Chen, C-N Wang. On the range edge of bird-plant specialized relationships: pollination by generalist passerines in *Aeschynanthus acuminatus* (Gesneriaceae) in Taiwan. Evolution meeting 2019: 21-25 June, Providence, RI, USA.

[3] **J-Y Lu**, K-S Chen, C-N Wang. Highly efficient pollination system in *Aeschynanthus acuminatus* (Gesneriaceae) reveals the potential role of generalized passerines in floral evolution of East Asia. XIX International Botanical Congress: 23-29 July, Shenzhen, China.

[4] **J-Y Lu**, K-T Hsin, C-N Wang. Gene duplication and relax from selective constraints of *Gcyc* genes create high floral diversity in Didymocarpoideae (Gesneriaceae). Society of Molecular Biology and Evolution Conference 2016: 3-7 July, Gold Coast, Queensland.

Outreach and Volunteer Experience

- **Rice Native Gardens Biodiversity Documentation Project** *2022–Present*
  Field Museum of Natural History, Chicago, USA [🌐]
  - Active contributor to biodiversity inventories of the Rice Native Gardens through iNaturalist.
  - Documented plant–pollinator interactions and seasonal biodiversity patterns.
- **CWA Cafe: Obsessed with Observations - Become an iNaturalist Super User** *2026*
  Chicago Wilderness Alliance, Chicago, USA [🌐]
  - Chicagoland City Nature Challenge series event. With Will Yaeger (University of Chicago) and Nicole Machuca (Field Museum).
- **73rd Member's Night** *2024*
  Field Museum of Natural History, Chicago, USA [🌐]
- **72nd Member's Night** *2023*
  Field Museum of Natural History, Chicago, USA
- **Bird Count at MSI** *October 2022*
  Museum of Science and Industry, Chicago, USA
- **Member's Virtual Event** *April 2022*
  Field Museum of Natural History, Chicago, USA
  - Lecture: Beyond hummingbirds
- **Herbarium Volunteer** *2016-2017*
  University of Melbourne Herbarium (MELU), Australia
- **Indigenous Tribe Outreach Program** *2012-2017*
  Digital and Humanities Caring Association, Taiwan [🌐]
  - Co-founder and volunteer mentor of school chapter at Taipei Municipal Chien Kuo High School (CK-ICTribe)
  - Student volunteer program connecting high school students in cities with elementary schools in indigenous communities across Taiwan, focusing on local culture, environmental knowledge, and community activities.

Professional Activities

Journal Peer Review

- ***Cladistics***
- ***Plant Species Biology***

Society Membership

- **Botanical Society of America**
- **American Society of Naturalists**

Field Work

Fieldwork forms a central component of my research on phylogeography and plant–pollinator interactions. I am deeply grateful for the support and hospitality of local communities who assisted with field access, logistics, and knowledge of local environments. My field sites are located on lands traditionally stewarded by indigenous peoples in Taiwan and by ethnic minority communities in southern China and the Central Highlands of Vietnam.

Regions of field research

- **Taiwan** *2016-2021; 2023-2025*
  New Taipei City; Nantou, Yilan, Chiayi counties; Central Mountain Range and more
- **Southern China** *2019-2021; 2023*
  Yunnan, Sichuan; Guangxi, Guangdong; Fujian
- **Vietnam** *2019; 2023*
  Central Highlands: Lam Dong Province